

Rocky Klopfenstein

702-600-9658 | rklopfenstein27@amherst.edu | linkedin.com/in/rockyklop | github.com/rklop

EDUCATION

Amherst College

Expected May 2027

Bachelor of Arts in Computer Science & Mathematics

Amherst, MA

- **GPA:** 4.00/4.00 | **Major(s) GPA:** 4.00/4.00 | **Relevant Coursework:** DSA, NLP, ML, AI Safety, Networks
- **On-Campus Roles:** COSC-111 Evening/Lab TA, CS Department Peer Tutor, Linear Algebra w/ Apps TA

EXPERIENCE

AI/ML Student Researcher

May 2025 – Present

Amherst College/VMware (In Collaboration)

Remote

- **Poster Presentation** at NeurIPS: 5th Math-AI Workshop — **ACCEPTED** to ICLR 2026: [Arxiv Paper Link](#)
- Investigated **Text-to-SQL**, leveraging logical equivalence checking to improve model accuracy and reliability
- Executed 3,500+ parallel verification jobs on High Performance Clusters, finding errors in benchmark verification
- Contributed to the open-source **VeriEQL** verification framework, extending symbolic SQL query evaluation
- Served as first author on a research paper with senior VMware researchers, driving experimentation and analysis
- *Currently* developing **Text-2-Code** methods to improve **LLM-generated code** leveraging formal verification
- Built AI agents for automated testing and C code-generation pipelines; designed and deployed MTurk surveys

AI/ML Software Engineering Intern

February 2025 – May 2025

PeakSpan Capital

New York, N.Y.

- Deployed a robust AI tool to filter novel B2B tech concepts in a quarterly-updated market intelligence database
- Designed a three-layer pipeline implementing statistical frequency filters, regex, and GPT-4o mini classification
- Reduced concepts from 470K to 80K, achieving a **83% compression rate** while preserving high-signal concepts
- Enhanced market-facing outputs, such weekly newsletters, boosting analyst productivity and insight quality

NLP Research Assistant

January 2025 – May 2025

Amherst College Natural Language Processing Lab

Amherst, MA

- Explored Universal Compositional Semantics (UDS) as a structural prior for **factuality detection** in LLMs
- Finetuned FACTKB, a factuality-detection framework, using regression/classification pipelines in Hugging Face
- Used the Amherst College High-Performance Computing Cluster to train LLMs on A100s, scheduling w/ Slurm

Founding Software Engineer

September 2024 – May 2025

Ideas 2 Innovations (I2I)

Amherst, MA

- Worked with student teams, engineering products for Amherst startups through a campus-led venture accelerator
- Built backend systems with LangChain, Django, and Google APIs, using Pandas and NumPy for data-driven tools
- Collaborated with cross-functional teams to design and deploy projects used by over **300 Amherst students**

PROJECTS

SmartWrite | *Python, React, Firebase, Vercel, WebSockets, OAuth2, Google APIs, Gemini* April 2025 – June 2025

- Built a Chrome Extension inspired by Cursor that integrates AI-powered writing tools directly into Google Docs
- Enables users to rewrite text, generate synonyms, and continue existing writing with context-aware completions
- Implemented secure Google sign-in via Firebase Auth, connected to a FastAPI backend using Gemini 2.5 Flash

Drop AI | *Python, Flask, RESTful APIs, Selenium, BeautifulSoup, OpenAI* September 2024 – November 2024

- Built a full-stack AI assistant that autofills attached PDFs and drafts contextual responses, **2nd at HackMIT**
- Customized for Relyvin, a medical-tourism startup, improving client communication and boosting email efficiency
- Integrated GPT-4 in a Flask backend for automated email categorization, summarization, and response use-cases

TECHNICAL SKILLS

Languages: Python, Java, SQL, JavaScript, TypeScript, C/C++, HTML/CSS, Bash/Zsh

Data Science & ML: HuggingFace, TensorFlow, PyTorch, LangChain, Pandas, NumPy, sklearn, Matplotlib, Seaborn

Backend: Node.js, Django, Flask, SocketIO, REST API, AWS, OpenXML SDK | **Frontend:** React.js, Next.js

Developer Tools: Git, Colab, Docker, Kubernetes, Github, Jupyter Notebook, GitLab, Cursor